Guidance and staff card

At check-out, record E for … against the pupil's target.

At check-out, record E for using the cell-cycle sequence to repair errors and explain matching chromosome sets against the pupil's target.

Prepared 3 November 2026

Invite a response. Give thinking time. The learner may pass. Offer private answers and the usual response mode.

Return to learning: use the individual pause plan, give space, then return to the last secure step and one structured choice. Correct privately.

Use soft instrumental music for transitions if the class plan says so.

Code	Meaning
VF	Verbal feedback given here
WS	Worked with support
I	Independent
NS + …	Next step one specific thing
E	Evidence captured on the evidence app
R	Responded loop closed after a pupil response and adult engagement
//	Self-edit point
?	Read this back to me

At check-out, record E for today's lesson against each pupil's target on the evidence app

Use observed evidence and record the support given. An edit is in the same colour, on the same page. Written feedback is used for extended writing when it adds value.

Codes follow the verified feedback policy, Issue 1. A button, saved answer or completed stage does not award a code.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt

Teacher and cover plan

At check-out, record E for using the cell-cycle sequence to repair errors and explain matching chromosome sets against the pupil's target.

I am learning to use the cell-cycle sequence to find and repair errors and explain matching chromosome sets.

Source sequence retained: Arrival 5; Starter 3; modelling 5 split across the two I do stages; first We do 7; second We do 5; You do 8; Review 4; Exit 3. Total 40 minutes.

Week arc: Discover: the cell cycle. Use: mitosis in sequence. Master: matching chromosome sets and repair.

Kit: large prepared paper chromosome strips, two cell outlines and a tray. Use the live lesson’s original copy, separate and split model, full and faded sequences, number hinge and two-fault challenge.

Keep the three settled reset opportunities inside shared work, independent work and review. Use an agreed pause and one-step return; add no separate stage.

Teaching sequence

Stage	Minutes	Teaching and check
Arrival	5	Q1 checks picture access. Compare the first and last pictures. Offer enlarged or tactile paper shapes; accept pointing before speaking.
Starter	3	Ask the original three retrieval questions in order. Invite a response and allow a pass. Reveal answers separately and revisit the success criteria.
I do	3	Show the complete original model. Step or play the six frames, then pause: is DNA copied before, during or after mitosis? Answer: before, in interphase.
We do	7	Use two quiet partner turns before writing. Hear the first wrong step and a reason. In Fault 1, repair interphase by copying DNA before mitosis. In Fault 2, repair anaphase by sending one copy of each type to each end. Check both repaired sets. Respond to the optional participation request by making or explaining a change and checking its effect.
I do	2	Model the six-chromosome case. Separate nuclear division from the final split. Ask what each new nucleus must receive.

Stage	Minutes	Teaching and check
We do	5	Collect simultaneous whiteboard, symbol or pointing responses and hear separate reasons. About three in four secure: move on. Otherwise re-model with two blue and two orange paper strips, then use the eight-chromosome parallel hinge. Correct answer: eight. Do not let a group result replace an individual response.
You do	8	Use the original ordering, matching-set and two-fault content through Supported, Independent and Aspire; Choice offers selected answers plus a reason. Give access support without dictating the science. Record actual independence and prompt level.
Review and improve	4	Give one specific feedback point. Wait for a real pupil edit or explanation and engage with it. Record only feedback that occurred.
Exit ticket	3	Hear every response through the usual mode. Use it to change the next example or prompt. Keep one response and state the support actually used.

Stage timings

Stage	Minutes
Today I can	0
Arrival	5
Starter	3
I do	3
We do	7
I do	2
We do	5
You do	8
Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Made by Matt

P17 · SLT planning and evaluation

Planned adaptations

Need	Planned adaptation
Cognition and learning	Keep the full six-phase strip beside the fault model. Compare one chromosome type first. Fade a phase-name cue only after a secure explanation.
Communication and interaction	Two quiet partner turns and the same speaking frame before writing. Accept point, sign or an exact scribe.
Regulation	Check access before the hinge and independent task. Use the agreed pause and one-step return privately.
Sensory and physical	Enlarged, high-contrast, easy-grip chromosome strips. Offer seated or standing handling and adult-directed placement.
Language access	Pre-teach the science terms with diagram and example. Allow planning in any language at the same scientific challenge.

If behind

Preserve copy, separate and split, each learner’s matching-set explanation, review and the two exit checks. After a secure hinge, shorten repeated oral rehearsal. Record the change and coverage impact. Use the next lesson, Master: matching chromosome sets and repair, to revisit an insecure phase cue.

Where catch-up is embedded

Arrival begins with the same labelled picture. The full sequence is available beside the faded one. Reference A and B stay beside You do. The catch-up sheet replaces a starting step after absence or an access barrier only.

Staff evaluation

Prompt	Record after teaching
Delivery	Delivered as planned, partly or changed; actual reason.
Criteria	Criterion, response and support given.
Adaptation	Actual adjustment and effect on a fresh response.
Coverage	What remains and the next teaching move.
Evidence	Location on the evidence app and this lesson code.

P18 · Individual target opportunities

Use opportunities from each learner’s existing plan. Keep pupil codes and target fields blank.

Opportunity	Lesson evidence and adult response
Cognition and learning	Find the first faulty step and demonstrate one independent repair. Record the pupil’s own sequencing, attention or independence target and actual prompt level.
Communication	Respond to the six-chromosome hinge in the agreed mode. Hear the learner’s meaning without requiring speech.
Regulation	Use the predictable picture start or agreed pause and re-entry step. Evidence the pupil’s own target privately.
Sensory or physical access	Direct or handle the model pieces using agreed access equipment. Separate access from science content support.
Language access	Use the actual vocabulary in the example; allow first-language planning and oral rehearsal. Record an agreed language target where relevant.

One target-matched evidence record can serve the existing plans. Task completion alone does not prove a target. Follow the individual support plan where it differs from these routines.

Made by Matt

Adult deployment and responsive feedback

Agree who supports the independent stage and who holds the group. With two adults, swap roles as useful. With one adult, use a prepared reference task while hearing learners separately.

Prompt least first: wait; point to the last complete step; ask what happened to the copied chromosomes; model one move with one type; let the learner try a fresh move with the other type. Use no physical prompting.

At I do: “Watch the DNA copy. Point to where the two matching sets go.” At We do: “Which is the first wrong step? What would change if you repaired it?” At You do: “Show me the last step you know. What does the model tell you next?”

Record the actual scientific response and content prompt, separately from the access aid. At handover, share what was independent, the feedback and the next step. Do not substitute a group product for an individual response.

Within the optional participation prompt, the lesson teacher hears the request, says what can change, makes one agreed change if possible, explains a limit and checks the effect.

Made by Matt

Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L2

First lesson back · once-only read

Before the lesson · scan the books

Scan for five minutes. Find where teaching stopped, existing feedback and actual pupil responses. Look for access barriers and useful science evidence.

Existing NS + is history

Do not chase, back-mark or retrospectively close earlier feedback. Begin fresh feedback from this lesson.

Ask the cover teacher and TAs

Ask what was taught, which model was used, which phase cues were faded and what changed for individual learners.

Check the roll

Compare attendance with the available work. New arrivals start from what they show here; absence creates no automatic deficit label.

Arrival read-off · decide by about minute eight

Read picture access before judging science. If number and matching sets are insecure, use large labelled outlines. If secure, fade a phase cue and ask for a reason.

After last lesson · one read, once

Read each criterion against the actual response and support given. Choose the next example or route from the least secure dependency. Then teach; do not reopen historical feedback.

Made by Matt

Preparation and practical safety

This is optional model rehearsal, not claimed core practical completion. Use large prepared paper chromosome strips, two cell outlines and a tray.

Loose strips could fall or be mouthed. Keep them on a desk or tray, out of mouths, and collect dropped pieces. Confirm the exact kit, local risk assessment and individual plans before use.

Offer enlarged print, tactile paper shapes, a static sequence, high contrast and adult-directed handling. Test pause, reset, keyboard controls and the offline display. Provide the same model on paper.

Destination link: a laboratory technician labels a cell sample and records what its microscope image shows. Keep model symbols distinct from microscope observations.

Made by Matt

Teacher answers and topic guidance

Accept scientifically equivalent answers in the learner's usual mode. Award each distinct point once. Keep answers hidden during independent attempts.

Starter

1. Active transport.
2. Water.
3. The nucleus.

Full and faded sequence

Interphase; prophase; metaphase; anaphase; telophase; cytokinesis. The faded step is anaphase: the two copies of each chromosome are pulled to opposite ends. DNA is copied before mitosis, in interphase.

Spot the mitosis error

Fault 1: the first error is in interphase. Copy DNA before mitosis so both new nuclei can receive matching sets. Fault 2: the first error is in anaphase. Send one A copy to each end, with a B copy at each end. The repair view shows the resulting matching sets after the later final split. Switching cases resets the previous diagnosis and repair.

Six-chromosome hinge

C: six in each daughter, with the same set as the original body cell. A, three: confuses this ordinary body-cell model with chromosome halving. B, twelve: treats copied material as the number retained by each daughter. D, none: leaves the copies in one cell instead of distributing them.

About three in four secure responses: move on. Otherwise re-model with two blue and two orange paper strips, labelled A and B, then use the parallel case. A starting set of eight gives eight in each daughter.

Transfer to repair

DNA is copied before matching copies separate into two nuclei. After the cell splits, matching new body cells can replace damaged cells. Accept the same explanation through a model demonstration.

Science scope and misconceptions

Mitosis separates copied chromosomes into two nuclei. Cytokinesis splits the cell. The model concerns ordinary body-cell division. It does not teach the full molecular detail, asexual reproduction or chromosome halving. Do not describe mitosis alone as the whole final split.

Copy before separation: both new nuclei need a matching set. Uneven separation can leave one nucleus with an extra copy and another lacking it. New matching body cells support growth and replacement of damaged cells.

Arrival Supported

1. Two cells.
2. The one cell on the left.
3. The inner outline labelled nucleus.
4. One cell has become two; each new cell has a matching labelled set.

Arrival Independent

1. Two.
2. The nucleus.
3. Both cells have A and B: the sets match.
4. One cell became two, each with a matching chromosome set.

Arrival Aspire

1. Two separate cells.
2. The nucleus.
3. Each new cell has both types, A and B, so the sets match.
4. It shows one cell becoming two with matching sets; it does not show the full molecular detail or every real chromosome.

Exit Supported

1. Before mitosis, in interphase.
2. Anaphase.

Exit Independent

1. Anaphase, when copies separate.
2. Send one A copy and one B copy to each end so each new nucleus receives both types.

Exit Aspire

1. Repair copying in interphase first. The copies must exist before the later separation can work.
2. Splitting the cell does not redistribute chromosome copies. Repair their separation into matching sets before the final split.

Made by Matt

Mark scheme Supported

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, mitosis and cytokinesis. Show A and B in each new cell. [3]

Interphase, mitosis, cytokinesis; matching A and B sets in both new cells. Award one point for each correctly placed stage, with matching sets checked as the success criterion.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Complete the faded sequence: prophase, metaphase, ___, telophase. Say what the missing step does. [2]

Anaphase; the two copies of each chromosome move to opposite ends.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 1 skips DNA copying. Choose interphase or cytokinesis as the first step to repair. Explain why. [3]

Interphase; copy DNA before mitosis; both new nuclei need a matching set of copies.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Independent

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Order interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Mark the four phases that belong to mitosis. [4]

Interphase first; prophase then metaphase; anaphase then telophase; cytokinesis last and the four middle phases identified as mitosis.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Fault 1 goes into mitosis without copying DNA. Find the first wrong step and explain its repair. [3]

First error in interphase; copy DNA before mitosis; this supplies matching copies for both new nuclei.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 2 sends both A copies to one new nucleus. Find the first wrong step, repair it and explain the result. [3]

First error in anaphase; send one A copy to each end; each new nucleus then receives a matching A and B set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Aspire

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]

Identify copying skipped in interphase; add the copying step before mitosis; identify both copies sent to one nucleus during separation; send one copy of each chromosome to each new nucleus before the final split.

Maximum 4 marks. Accept equivalent valid wording or a clear model demonstration.

2. Explain why swapping mitosis and DNA copying would break the model. [3]

DNA copying belongs before mitosis; without copies there is not a copied set to distribute to each new nucleus; both new cells would not receive the intended matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

3. Predict what would happen to the sets if copied chromosomes did not separate evenly. Give a reason. [3]

New nuclei could have unequal sets; one could gain a chromosome copy and another lack it; the sets would no longer match because separation was uneven.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Mark scheme Choice

Use the actual response and support given. These question tariffs are not participation scores. Award each distinct science point once, up to the stated maximum.

1. Choose the first, middle and last stage from interphase, mitosis and cytokinesis. Show both A and B in each new cell. [3]

Interphase first, mitosis middle, cytokinesis last, with matching labelled sets checked.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

2. Choose anaphase or telophase for the missing step after metaphase. Show what happens to the copies. [2]

Anaphase; the two copies of each chromosome move to opposite ends.

Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.

3. Fault 1 skips DNA copying. Choose copy DNA first or split the cell first as the repair. Give a reason. [3]

Copy DNA first; copying belongs in interphase before mitosis; each new nucleus needs a matching set.

Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.

Made by Matt

Science staff guidance

Invite a response, give thinking time and allow a pass. No learner is required to read difficult text aloud. Accept point, sign, speech, drawing, exact scribing and adult-directed handling.

Tell the teacher the barrier observed: reading, vocabulary, processing, regulation, writing or science. Avoid prompting through every question or supplying the final answer.

First check: DNA is copied before mitosis. Second check: both new nuclei receive a matching set. Third check: the cell still needs the final split. Re-model only the insecure step, then use a fresh phase or number case.

The teacher engages with an actual review response and uses the exit to change the next model or prompt. Use one usual-mode response as evidence. Keep the individual support plan in charge of regulation and access.

Made by Matt

Science · LAUNCH · Autumn 2 · Week 1 · SCI-LAUNCH-A2-W01-L2

TA Brief

At check-out, record E for using the cell-cycle sequence to repair errors and explain matching chromosome sets against the pupil's target.

I am learning to use the cell-cycle sequence to find and repair errors and explain matching chromosome sets.

Key checks: copy DNA in interphase; separate matching chromosome copies during mitosis; split the cell in cytokinesis. The four phases of mitosis are prophase, metaphase, anaphase and telophase.

Offer the full sequence beside the faulty model. Diagnose interphase in Fault 1 and anaphase in Fault 2. Wait, gesture, rephrase, then model one type before a fresh pupil move. Give time for quiet partner rehearsal before writing.

Hinge: six chromosomes in each daughter. Parallel check: eight in each daughter. Hear the reason and use the full answer guide for a misconception diagnosis.

At You do, record actual I · S · C beside the task. Access support and content prompts are recorded separately. Use a quiet pause and one-step return in line with the individual plan.

Report what the learner showed, the barrier, the support and the next move. A correct click alone is not an explanation.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
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Exit pupil voice
What helped you most today, or what would you change next time?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Stage timings

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Arrival	5
Starter	3
I do	3
We do	7
I do	2
We do	5
You do	8
Review and improve	4
Exit ticket	3
Tools & print	0

Total: 40 minutes

Made by Matt